

PathFinder – AI-Powered Internship & Fellowship Platform

1. Project Overview

1.1 Introduction

Finding the right internship or fellowship is often a fragmented and overwhelming process for students. Simultaneously, organizations struggle with managing high volumes of applicants efficiently without missing top talent. **PathFinder** is a modern, AI-powered platform designed to bridge this gap. It connects students with high-quality opportunities while providing organizations with an intelligent Applicant Tracking System (ATS). By deeply integrating Google Generative AI (Gemini), PathFinder automates tedious tasks such as resume parsing, profile creation, cover letter generation, and candidate evaluation, creating a seamless, centralized experience for all stakeholders.

2. Business Problem

Most organizations and students use disconnected systems to manage applications and hiring. Current challenges include:

- Students spend excessive time manually entering profile details across different platforms.
- Writing customized Statements of Purpose (SOPs) or cover letters for every application is time-consuming.
- Organizations manually screen hundreds of resumes without intelligent filtering.
- Applicant tracking is often handled via disjointed email threads and spreadsheets.
- Slow approval and hiring processes lead to losing top candidates.
- Lack of automated feedback and transparent application tracking for students.

These problems result in delayed decision-making, reduced productivity, and a poor experience for both applicants and recruiters.

3. Proposed Solution

- Develop a centralized PathFinder Platform that provides:
- AI Resume Autofill (Profile Generation)
- AI SOP & Cover Letter Generator
- Centralized Opportunity Discovery (Search & Bookmarking)
- Intelligent Applicant Tracking System (ATS)
- AI Candidate Match Scoring & Summaries
- Document Management (Resumes, Transcripts)
- Analytics & Admin Dashboard
- Real-Time Notifications
- Secure Role-Based Access Control

4. Project Objectives

Objective ID	Objective
OBJ-01	Automate student profile creation using AI resume parsing
OBJ-02	Digitize and streamline the internship application process
OBJ-03	Reduce manual screening time for organizations using AI Match Scoring
OBJ-04	Improve candidate quality through AI-assisted SOP generation
OBJ-05	Provide centralized document storage for students
OBJ-06	Implement a transparent Applicant Tracking System (ATS)
OBJ-07	Ensure secure role-based access for Students, Orgs, and Admins
OBJ-08	Build a responsive, user-friendly frontend with a modern aesthetic

5. Expected Benefits

Benefit	Description
Reduced Manual Work	Eliminate manual data entry for student profiles and resume screening.
Improved Accuracy	AI extraction ensures skills and experiences are accurately categorized.
Faster Hiring	ATS workflows speed up candidate progression (Reviewing -> Shortlisted -> Accepted).
Better Communication	Automated status update notifications keep applicants informed.
Centralized Data	A single source of truth for all opportunities and applications.

6. Project Scope

In Scope

The project will include the following modules:

Module	Included
Authentication (JWT)	✓
Role Management	✓
Student Profiles	✓
Organization Profiles	✓
Opportunity Management	✓
Application & ATS	✓
Document Management	✓
AI Resume Autofill	✓
AI SOP Generator	✓
AI Candidate Scoring	✓
Notifications	✓
Admin Dashboard	✓

Out of Scope

The following features are excluded from this version:

- Native Mobile Application
- Built-in Video Conferencing / Interviewing

- Payment Gateways (for paid opportunities)
- External Job Board Integrations (LinkedIn/Indeed sync)
- Multi-language Support
- Offline Functionality

7. Stakeholders

Stakeholder	Responsibility
System Administrator	Overall platform moderation, managing spam users and inappropriate posts.
Organization HR / Recruiter	Post opportunities, review AI match scores, and progress candidates through the ATS.
Student / Applicant	Upload documents, generate AI content, apply to opportunities, and track statuses.

8. User Roles

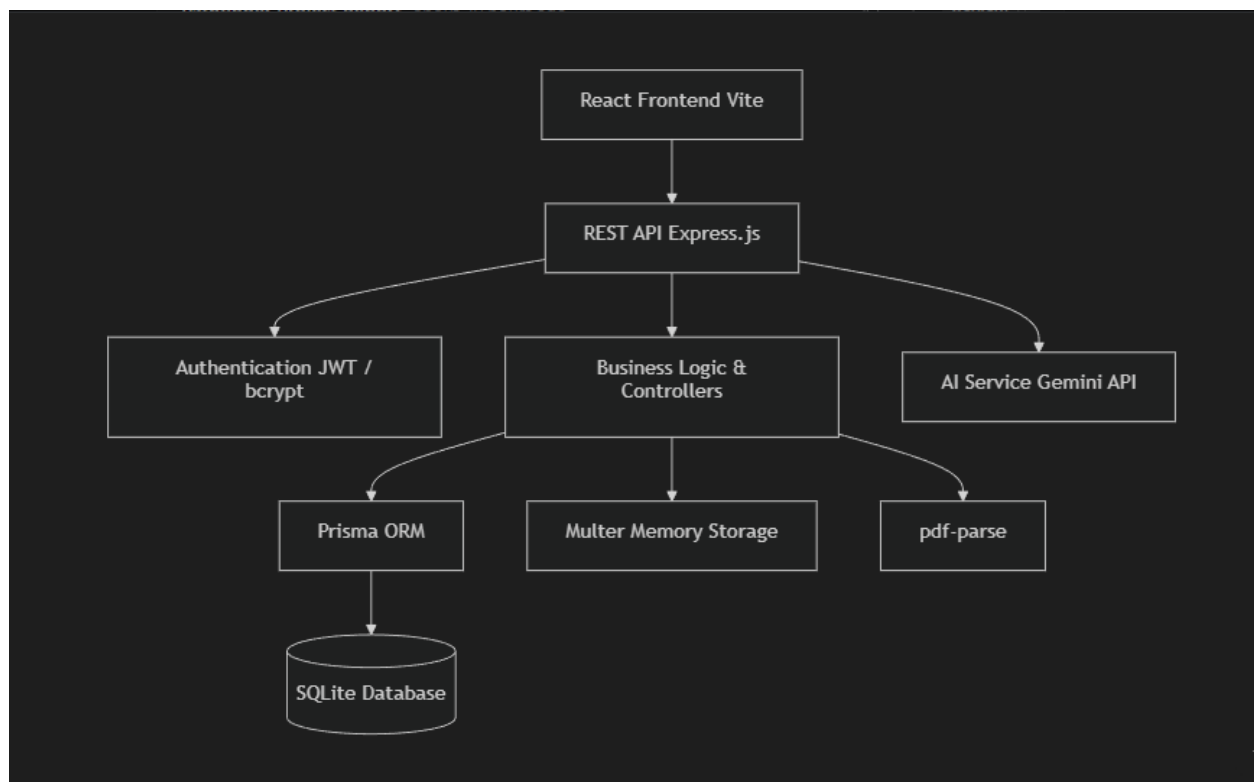
Role	Primary Responsibility
Admin	Complete control over the platform health and global moderation.
Organization	Recruitment lifecycle management for their specific company.
Student	Personal profile management and job application processing.

9. Role-Based Access Control (RBAC)

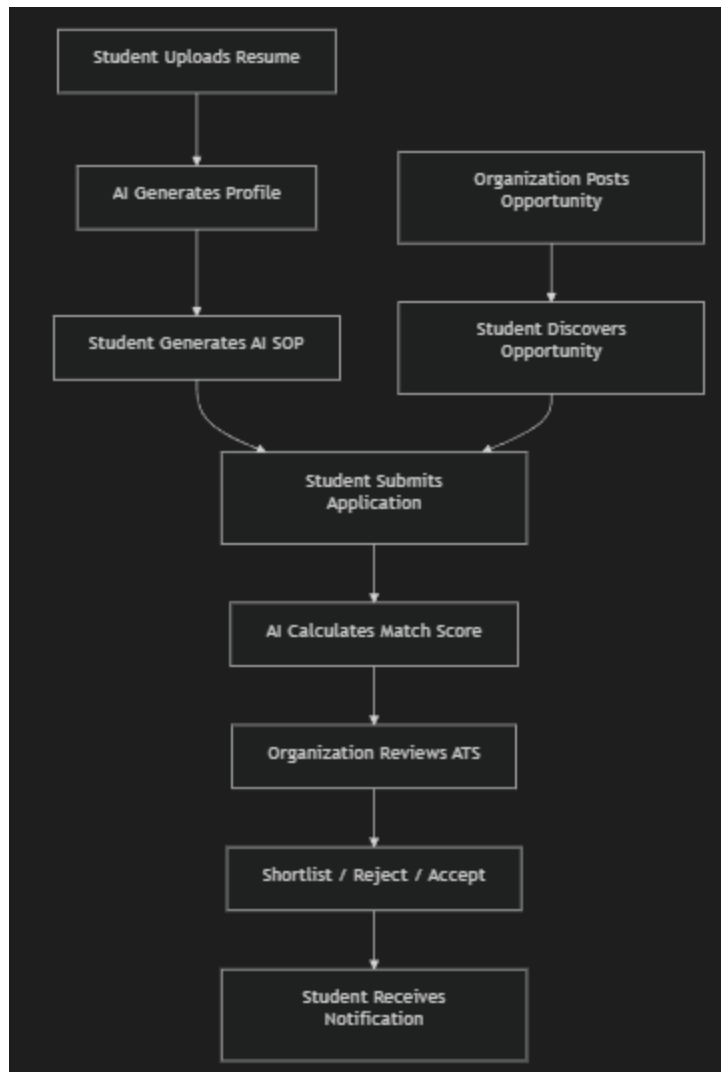
Module	Admin	Organization	Student
User Management	Full	No	No
Organization Profile	Read	CRUD (Own)	Read
Student Profile	Read	Read (Applicants only)	CRUD (Own)

Module	Admin	Organization	Student
Opportunities	CRUD (Global)	CRUD (Own)	Read / Bookmark
Applications (ATS)	Read	CRUD (Status)	Create / Withdraw / Track
AI Generators	No	Candidate Summaries	Resume Parse / SOP Gen

10. High-Level System Architecture



11. Core Business Workflow



12. Key Functional Modules

Module ID	Module Name	Description
M-01	Authentication	User registration, login, JWT security
M-02	Profile Management	Student and Org profiles, skills, portfolios
M-03	Document Management	Secure resume and transcript uploads
M-04	Opportunity Engine	Drafting, publishing, and archiving opportunities
M-05	Application (ATS)	Applicant tracking and

Module ID	Module Name	Description
M-06	AI Assistant	status management Resume parsing, matching, and generation
M-07	Notifications	Real-time in-app alerts
M-08	Admin Moderation	Global data oversight

13. AI Assistant Integration

The AI Assistant (powered by Google Gemini) is integrated contextually based on user roles:

Feature	Description
Resume Analyzer	Extracts Bio, Skills, Major, and University from a PDF resume to instantly build a profile.
SOP Generator	Crafts a tailored Statement of Purpose by mapping student skills to specific opportunity requirements.
Match Scoring	Automatically evaluates an applicant's cover letter and profile against the job description, returning a 0-100% score.
Candidate Summarizer	Generates a 2-3 sentence summary of a candidate for quick recruiter screening.

14. Assumptions

- Users will access the application through a modern web browser.
- External AI services (Gemini) are online and responsive.
- SQLite is sufficient for local development and initial deployment.
- Resumes uploaded are in valid, text-readable PDF format.

15. Constraints

Constraint	Description
Technology	PERN/SERN Stack (PostgreSQL/SQLite, Express, React, Node) only.

Constraint	Description
AI Rate Limits	External AI API usage is subject to Gemini rate limits.
Storage	File uploads are currently handled in memory or local disk; requires cloud storage (e.g., AWS S3, Cloudinary) for scale.

16. Success Criteria

The project will be considered successful if:

- Role-based access control successfully routes Students, Orgs, and Admins to their respective dashboards.
- AI successfully extracts accurate information from at least 90% of standard PDF resumes.
- Organizations can seamlessly progress candidates through the ATS stages.
- The UI reflects the required premium Dark Mode and Glassmorphism aesthetic.
- All core modules are integrated without breaking changes.

17. API Architecture & Endpoints

The platform utilizes a modular REST API built with Express.js. The endpoints are securely grouped by logical domains:

17.1 Core Endpoints

- **/api/auth:** Handles user authentication, including registration, login, and password reset functionalities. Employs JWT security and rate limiting.
- **/api/student:** Manages student profile creation, skill tagging, certifications, portfolio links, and document retrieval.
- **/api/org:** Manages organization profiles, branding (logos), and company details.
- **/api/admin:** Secured routes for platform moderation, audit logs, and global data management.

17.2 Opportunity & Application Endpoints

- **/api/opportunities:** CRUD operations for internships, scholarships, and grants. Supports status management (Draft/Published) and category filtering.

- **/api/applications:** Manages the core ATS workflow. Handles application submissions, status updates by organizations, withdrawal by students, and tracking timeline events.
- **/api/search:** Dedicated endpoints for complex search queries and retrieving user's bookmarked opportunities.

17.3 Utility & Integration Endpoints

- **/api/ai:** Interfaces with the Google Gemini AI service. Exposes endpoints for resume parsing, tailored SOP generation, candidate summarization, and AI match scoring.
 - **/api/notifications:** Retrieves user-specific notifications, handles read/unread status updates, and fetches unread counts for UI badges.
 - **/api/stats:** Exposes aggregate platform data for public display (e.g., active student count, open opportunity count).
 - **/api/upload & /api/files:** Handles secure file uploads for resumes, transcripts, and organization logos using Multer, as well as role-based protected file serving.
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Part 2 – Functional Requirements (Core Modules)

Module 1 – Authentication & User Management

1.1 Module Overview

Authentication ensures secure entry into the platform. Users register under specific roles (STUDENT, ORGANIZATION, ADMIN) which strictly dictate their permissions and routing.

Business Objective

Protect user data while providing a seamless onboarding experience.

Business Rules

- Emails must be unique across the platform.
- Passwords must be securely hashed using bcrypt.
- A valid JWT token is required for protected routes.
- Unknown roles default to immediate rejection.

Acceptance Criteria

- ☑ User can register securely.
- ☑ Passwords are encrypted in the database.
- ☑ Invalid logins return generic error messages to prevent enumeration.
- ☑ JWT is issued and attached to subsequent API calls.

Module 2 – AI Application & ATS Management

2.1 Module Overview

This module handles the core interaction between Students and Organizations. It manages the submission of applications and the subsequent review process.

Features

- Submit Application with Cover Letter
- AI Match Score Calculation
- Status Updates (Pending -> Reviewing -> Shortlisted -> Accepted/Rejected)
- Undo Status Change
- Application Timeline Tracking (ApplicationEvent)

Application JSON Model

```
{
  "id": "APP_1023",
  "opportunityId": "OPP_001",
  "studentId": "STU_045",
  "status": "SHORTLISTED",
  "matchScore": 88,
  "matchReason": "Strong React skills align with requirements.",
  "coverLetter": "I am writing to express my interest...",
  "events": [
    { "type": "SUBMITTED", "date": "2026-07-01" },
    { "type": "STATUS_CHANGED", "details": "Moved to Shortlisted", "date": "2026-07-03" }
  ]
}
```

Acceptance Criteria

- ☑ Students can apply to opportunities.
- ☑ AI automatically generates a match score upon submission.
- ☑ Organizations can change candidate statuses.

- ☒ Notifications are fired to the student on status change.
- ☒ Application history is preserved via events.

Module 3 – Platform Features

3.1 Notifications

PathFinder includes a robust notification system designed to keep users informed about important events.

- **Real-Time Alerts:** Students are notified immediately when their application status changes (e.g., Shortlisted, Rejected, Accepted).
- **Opportunity Alerts:** Notifications for approaching deadlines or when an application is withdrawn.
- **System Notifications:** Admin announcements and platform-wide alerts are supported.
- **Unread Tracking:** Users can see unread notification counts and mark notifications as read directly from the notification center.

3.2 Advanced Search & Bookmarks

- **Semantic Search:** Users can search for opportunities using keywords, categories (Internship, Scholarship, Hackathon), or locations.
- **Bookmarking:** Students can save opportunities to a "Bookmarks" list to apply later.
- **Saved Searches:** Users can save specific search criteria to easily find new matches in the future.

3.3 Analytics & Statistics

- **Public Stats:** The dynamic landing page displays real-time platform statistics such as the total number of active students, open opportunities, and top organizations.
- **Dashboard Metrics:** Students and organizations have access to quick metrics on their respective dashboards (e.g., total applications submitted, total candidates reviewed).